A-A-51039A
10 August 1990
SUPERSEDING
A-A-51039
21 November 1985

COMMERCIAL ITEM DESCRIPTION

ALUMINUM HYDROXIDE GEL AND MAGNESIUM CARBONATE SUSPENSION

The General Services Administration has authorized the use of this Commercial Item Description.

Salient Characteristics:

Shall be Aluminum Hydroxide Gel and Magnesium Carbonate Suspension.

Each 15 ml shall contain 95.3 mg of Aluminum Hydroxide, 412.2 mg of Magnesium Carbonate, Sodium Alginate, a suitable preservative, and other suitable ingredients acceptable to the FDA.

Each 15 ml shall contain not less than 87.0 percent and not more than 114.0 percent of the required amount of Aluminum Hydroxide when assayed by a suitable, accurate and reproducible assay method. Shall contain not less than 88.8 percent and not more than 111.4 percent of the required amount of Magnesium Carbonate when assayed by a suitable, accurate, and reproducible method.

Each 15 ml shall have an acid neutralizing capacity of not less than 5 mEq and shall result in a pH of 3.5 or greater at the end of the initial 10 minute period when tested using the Acid Neutralizing Capacity Test in the FDA Over-The-Counter Monograph.

Suspension shall contain not more than 100 aerobic microorganisms per milliliter when tested using a suitable, accurate, and reproducible method.

AMSC\NA FSC 6505

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Suspension shall meet the requirements for the test for absence of the organisms listed below under the USP Microbial Limits Tests (61):

Escherichia coli
Salmonella
Staphylococcus aureus
Pseudomonas aeruginosa
Pseudomonas fluorescens

Suspension shall be tested for the effectiveness of its preservative using the following Preservative Effectiveness Stability Test below or a similar suitable, accurate and reproducible method approved by the Food and Drug Administration. The samples shall pass the Preservative Effectiveness Stability Test if the initial inoculum count (NLT 10^3 organisms per ml of product after initial inoculation with an organism) does not increase by more than 50% by the third week and remains at or below that level by the fourth week.

Preservative Effectiveness Stability Test.

Preparation of Inoculum. Validate spectrophotometer used in this assay. Prior to the test, inoculate the specified number of Tryptic Soy Agar (TSA) slants (for the Staph Aureus and Enterobacter Cloacae) and Sabourand Dextrose Agar (SAB) slants (for Aspergillus niger) and incubate as follows:

Organism	No. of Slants	Incubation Conditions
Staph aureus ATCC 6538	3	30-35°C., 18-24 hours
Enterobacter cloacae ATCC 23355	3	30-35°C., 18-24 hours
Aspergillus niger ATCC 16404	6	20-25°C., 7 days

Procedure:

Sample Preparation

Prepare a set of four sterile sample containers (tubes or bottles). Label three of the tubes with product, lot number and organism. Label one container as a control and fill each container with 20 ml of the suspension.

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Inoculum Preparation

Prepare suspension of Staph aureus and Enterobacter cloacae by washing the growth on the TSA slant into 4-5 ml of sterile phosphate buffer. Transfer 1.0 ml into a sterile cuvette. Adjust turbidity of suspension to appropriate optical density and dilute as follow. A minimum of 1000 oraganism/ml of product is required, not exceeding 5000 organisms per ml of product as the upper limit.

Organism	Optical Density at 580 nm
Staph aureus	0.4
Enterobacter cloacae	0.2

Prepare the seed of Aspergillus niger by washing 6 SABS slants into 5 ml of phosphate buffer +0.05% of TWEEN 80 in a sterile 16 x 100 mm test tube. Inoculate 0.1 ml of 10^{-2} dilution into 20 ml of the product.

Inoculate 0.1 ml of 10^{-3} to 0.1 ml of 10^{-2} (depending upon which spectrophotometer is used) dilution each of Staph aureus and Enterobacter cloacae into two 20 ml samples of the aluminum hydroxide gel.

Sample Inoculation

Prepare seeds one at a time and complete all necessary dilutions prior to preparing next seed.

Verify inoculum by performing back/titer plates for each organism.

Dilute each organism to 10^{-7} in PO_4 and inoculate 2 plates per dilution with 0.1 ml of 10^{-5} -10^{-7} dilutions. Spread 0.1 ml inoculum over entire surface of plates using a sterile 10 ul inoculating loop or use 1 ml for pour plate. Incubate Enterobacter and Staph plates at 30-35°C for 24-48 hours and Aspergillus plates 20-25°C for 24-48 hours. Use TSA plates for Staph and Enterobacter and SABS plates for Aspergillus niger.

Pipet 0.1 ml of appropriate dilution of each organism into 20 ml of Aluminum Hydroxide Gel into labeled container. Mix well. Keep samples at room temperature for test interval.

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Test Interval

Seven days after initial inoculation of product, sample Aluminum Hydroxide Gel by pipetting 1.0 ml from original container into 9.0 ml of phosphate buffer and perform a solution series. Prepare duplicate pour plates at 10^{-1}, 10^{-3}, and 10^{-5} dilution by pipetting 1.0 ml samples from each dilution into two 15 x 100 mm petri dishes. Add 18-20 ml of appropriate liquified Agar and swirl to mix. Allow plates to solidify and invert and incubate at 30-35°C for at least 48 hours. Incubate Aspergillus niger at 20° - 25°C for at least seven days. Range of dilutions will have to be adjusted as bacterial populations increase or decrease. Repeat procedures in 4-a at 14, 21, and 28 day intervals.

Calculations

To determine the theoretical yield of 0.1 ml of each organism in 20 ml of Aluminum Hydroxide Gel, divide the CFU/ml of seed (obtained from count from backtiter plates performed from organism dilutions) by 200 (0.1 ml into 20 ml is a 1:200 dilution). This count should be NLT 10^3 organisms per ml of product after initial inoculation. Use this theoretical figure as a base to determine reduction during subsequent test intervals.

To determine the population in the product, multiply the average count of the two pour plates times the reciprocal of the dilution factor.

The test sample passes the Preservation Challenge test if the initial inoculum level does not increase by more than 50% by the third week, and remains at or below that level by the fourth week.

Workmanship. The suspension shall be free from defects which detract from its/their appearance or impair its/serviceability.

Unit. Bottle (BT).

For Size 1 - One bottle containing 12 fl oz (355 ml), as specified, constitutes one unit. Unit bottle shall have either an inner or an outer tamper-evident seal.

For Size 2 - One bottle containing 2 oz (60 ml), as specified, constitutes one unit. Unit bottle shall have either an inner or an outer tamper-evident seal.

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Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

NSN	ITEM IDENTIFICATION	SIZE
6505-01-083-6612	ALUMINUM HYDROXIDE GEL AND MAGNESIUM CARBONATE SUSPENSION, 12 fl oz (355 ml)	1
6505-01-249-6615	ALUMINUM HYDROXIDE GEL AND MAGNESIUM CARBONATE SUSPENSION, 2 oz (60 ml)	2

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Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:
DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITY:

VA-OSS
PHS
FDA-MPQAS

PGC 52077

Project No. 6505-2890

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-51039A	ALUMINUM HYDROXIDE GEL AND MAGNESIUM CARBONATE SUSPENSION

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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